

Month 1 - Python Fundamentals: The Building Blocks

Topics:

- Introduction to Python, installation, variables, data types (integers, floats, strings, booleans)
- Operators (arithmetic, comparison, logical)
- Input and Output
- Control Flow: `if`, `elif`, `else` statements

Project:

- A simple calculator that takes two numbers and an operator from the user and performs the calculation.

Resources:

- [Python for Everybody - Free Course](<https://www.py4e.com/lessons>)
- [Official Python Tutorial](<https://docs.python.org/3/tutorial/>)

Practice:

- [HackerRank - Python Basics](<https://www.hackerrank.com/domains/python/py-basic>)

Month 2 - Data Structures and Loops: Organizing Information

Topics:

- Lists, Tuples, Dictionaries, Sets
- `for` loops, `while` loops
- List comprehensions
- Basic string manipulation

Project:

- A to-do list application where users can add, view, and mark tasks as complete.

Resources:

- [Python Crash Course (Book - often available at libraries)](<https://nostarch.com/pythoncrashcourse>) (Focus on Ch 1-5)
- [Real Python: Python Data Structures](<https://realpython.com/python-data-structures/>)

Practice:

- [LeetCode - Easy Python Problems](<https://leetcode.com/problemset/all/?difficulty=EASY&page=1&topic=%5B%22Data%20Structures%22%5D>) (Focus on array/list manipulation)

Month 3 - Functions and Modules: Reusability and Organization

Topics:

- Defining and calling functions
- Function arguments (positional, keyword)
- Scope (local, global)
- Importing and using modules (like ``math``, ``random``)

Project:

- A random password generator that allows users to specify length and include/exclude certain character types.

Resources:

- [Python for Everybody - Functions](<https://www.py4e.com/lessons/functions>)

- [GeeksforGeeks: Python Functions](https://www.geeksforgeeks.org/python-functions/)

Practice:

- [Exercism.org - Python Track](https://exercism.org/tracks/python) (Focus on exercises requiring functions)

Month 4 - File Handling and Error Management: Interacting with the World

Topics:

- Reading from and writing to files (text files)
- Exception handling (`try`, `except`, `finally`)
- Common file operations
- Working with CSV files

Project:

- A simple contact book that saves/loads contacts from a CSV file.

Resources:

- [Python for Everybody - Files](https://www.py4e.com/lessons/files)
- [Real Python: Python File I/O](https://realpython.com/read-write-files-in-python/)

Practice:

- [Coderbyte - Python Challenges](https://coderbyte.com/challenges) (Look for challenges involving file operations or data processing)

Month 5 - Object-Oriented Programming (OOP): Building Complex Applications

Topics:

- Classes and objects
- Attributes and methods
- Inheritance and polymorphism
- Encapsulation

Project:

- A simple "Inventory Management System" for a small store, using classes for `Product` and `Inventory`.

Resources:

- [Python for Everybody - Classes](<https://www.py4e.com/lessons/oo>)
- [Real Python: Python Classes and Objects](<https://realpython.com/python-classes-objects/>)

Practice:

- [Codewars - Python Kata (search for OOP related tags)](<https://www.codewars.com/kata/search/python?q=oop&beta=false>)

Month 6 - Libraries for Job Readiness: Data Analysis and Web Interaction

Topics:

- Introduction to `NumPy` for numerical operations
- Introduction to `Pandas` for data manipulation
- Basic `Requests` for making HTTP requests (fetching data from APIs)
- Introduction to web scraping (e.g., using `BeautifulSoup`)

Project:

- Fetch data from a public API (e.g., a weather API or a joke API), process it with Pandas, and display key information.

Resources:

- [NumPy Official Documentation](https://numpy.org/doc/stable/)
- [Pandas Documentation](https://pandas.pydata.org/docs/)
- [Python Requests Documentation](https://docs.python-requests.org/en/latest/)
- [Beautiful Soup Documentation](https://www.crummy.com/software/BeautifulSoup/bs4/doc/)

Practice:

- [Kaggle - Python Notebooks (explore public datasets and code)](https://www.kaggle.com/code)
- [HackerRank - API exercises (if available, or practice with public APIs directly)](https://www.hackerrank.com/)